SAD - Ignite R3.0 Digital Offer Presentation

Solution Architecture for VH220041 - Ignite 3.0 Digital Offer Presentation

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Threat Risk Assessment:	

How to Complete this Document

How to Complete this Document

All typing in *blue italics* in the body of this document is for instruction and guidance and must be removed on completion of the document.

The document is structured into 7 sections

***Section 1** is the Context designed to understand the project objectives, scope, Business Context overview and critical success factors. This is required for both solution Outline and Solution architecture documents*

***Section 2** is the End to End solution description which provides solution options (if any), recommended solution rationale along with architectural overview of the recommended solution from contextual and components perspective. This section will be reviewed by the principal architects as part of architecture assessment of solutions to ascertain if the recommended solution aligns to Architecture policy and blueprints.*

***Section 3** provides the Non-functional overview section, to ensure that the recommended solution addresses the non-functional requirements of the respective areas including the design standards and standard operating environments.*

6.3 Model Building blocks (including testing tools)
7 Deployment/Migration Strategy
8 Future Considerations
9 Appendix

Section 4 *Security Legal and Regulatory Aspects of the Proposed Solution*

Section 5 *provides the description of the Non-production environment, the changes that are required in the environment, dependencies, assumptions and constraints*

Section 6 *provides deployment and migration strategy.*

Section 7 *provides future considerations.*

ARB presentation

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Document Control

Change History

Change History

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0.1	May 15, 2026	Draft	@Asim Rasheed
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		Serices and Domain Impacted	
0.3	May 26, 2026	Updated Solution Architecture Diagram Added Component Impacts Added detailed sections for Component Impacts Updated Mandatory 14 details	@Asim Rasheed
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Distribution

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References

References

Provide a list of all other documents to consider when reading this document

The following documents are referenced:

Reference	Document	Version

Definitions

Definitions

Definitions of all terms used in this document to define the solution (glossary)

Term / Abbreviation	Description / Definition
BRS	Business Requirement Specification
ETE	End to end

SO	Solution Outline
SAD	Solution Architecture Document
HLD	High level Design
LLD	Low level design or Detailed design
BB	Building block

Context

Required for both SO and SA

Purpose

This initiative is part of the Ignite 3.0 Release and is mainly covering the minimal scope for replacing current state RTDM offers in MyVF (OneApp + OneWeb) Dashboard experience.

The current RTDM offer retrieval will be replaced with Pega. Pega will act as the primary decisioning engine for this experience, responsible for evaluating eligibility, prioritisation, and suppression before returning a clear outcome to the digital channels. Digital channels will not perform decisioning or arbitration logic.

Business Benefit

Required for ARB presentation Next

Presenting personalised NBA offers increases personalization capabilities, omni-channel enablement, increase sales & revenue, decrease churn and create relevancy

Establishing this capability and integration will deliver significant operational and customer experience advantages, including:

1. **Streamlined Offer Management**
   - Enable CLM business teams and Digital & IT operational teams to configure digital templates for customer display.
   - Utilise AEM for image hosting and Pega for offer configuration and decisioning.
2. **Enhanced Digital Experience Control**
   - Allow Digital OneApp to manage the rendering of offers within the “Offers For You” section and other related experiences.

- Ensure Pega remains the authoritative source for all offer configuration and decision-making processes.

3. **Agile Offer Deployment**
   - Support creation and removal of offers based on customer eligibility and business rules, fully managed within Pega.
   - Eliminate the requirement to introduce offer logic within digital channels.
4. **Improved Offer Prioritisation**
   - Facilitate prioritisation of offers at User/Contact level across customer accounts or multiple services, leveraging models and business rules.
5. **Campaign Migration and Consolidation**
   - Migrate critical campaigns from RTDM to Pega, enabling a complete digital cutover and reducing dependency on legacy systems.
6. **Data Capture for Decisioning and Analytics**
   - Collect and utilise interaction data, including impressions and response feedback, from digital channels within CLM.
   - Support continuous analytics and learning without shifting decision-making responsibilities to digital channels.
7. **Business Continuity and Customer Experience Assurance**
   - Implement a defined fallback experience to maintain an uninterrupted digital customer experience in scenarios where:
     - No eligible offers exist.
     - Pega fails to respond within agreed performance thresholds.

Scope

Required for ARB presentation Next Back

In Scope

Release 3.0 introduces changes to MyVodafone to support the delivery of personalised offers through Pega Customer Decision Hub (CDH). The digital channels will retrieve and display offers and next-best actions provided by Pega within the "For You" experience, replacing the current postpaid offer presentation capability supported by SAS RTDM. As part of the transition, customer offer and campaign data from existing sources may be provided to Pega to support offer presentation and continuity of customer experience. The digital channels will also provide customer interaction feedback, such as offer impressions and responses, to support offer reporting, optimisation and decisioning outcomes.

- Integrate OneApp with Pega CDH to retrieve and display personalised offers and next-best actions.
- Transition offer decisioning from SAS RTDM to Pega, with Pega managing eligibility, prioritisation, suppression, and offer selection.
- Surface Pega-selected offers within the "Offers For You" experience while digital channels continue to manage presentation and UX.
- Capture and send customer engagement events (e.g., impressions, clicks, responses) back to Pega to support reporting, analytics, and optimisation.
- Enable personalised offer experiences using customer, account, and service-level context provided to Pega.
- Continue supporting prepaid offers by leveraging SAS RTDM metadata, with Pega determining applicable campaigns and offer presentation.
- Support migration of existing offer and campaign data to Pega to maintain continuity of customer experiences.
- Provide resilience and fallback experiences when no eligible offers are available or Pega services are unavailable, ensuring a consistent customer experience.

Out Of Scope

- Any Changes to Digital offer fulfilment, Pega will present similar offer set as RTDM, and fulfilment will be routed to assisted journey through CTAs
- Push Notifications, Active notifications where the customer is presented with a notification and must click the notification to proceed.

Business Context Overview

Required for ARB presentation Next Back

NBA offers are configurable with the ability to display individual offers. These offers are presented at configurable points throughout the My Vodafone session.

During a OneApp session, the offers relevant to the specific customer will be fed from Pega (replacing the current SAS RTDM). The customer's activity in OneApp, including the viewing of, take up or rejection of offers presented is recorded throughout the session and fed back to Pega.

Process	Impacted Business Unit/s	Description of change

OneApp Dashboard	Self-Serve Online, CVM	OneApp dashboard will host top 4 NBA offers
OneApp Activity	Self-Serve Online, CVM	NBA offer capability in OneApp activities
OneApp Offers page	Self-Serve Online, CVM	OneApp “Offers for You” page will contain all NBA offers

Products and Business Services

Required for ARB presentation Next Back

Contextual elements associated with the recommended solutions are as follows. Provide the brands, products and business services that are impacted by the solution.

Brands in Scope

Brand	In Scope(Y/N)	Comments
Vodafone	Y	Only include Vodafone offers on OneApp Web & App
TPG	N	Not in scope
iiNet	N	Not in scope
Lebara	N	Not in scope
Felix	N	Not in scope
AAPT/Internode	N	Not in scope
Other	N	Not in scope

Products

Include the list of products impacted with this solution. The product catalogue is available here.

Product(s)	Comments

Business Services

The list of Business Services can be accessed here. If the Business service is not in this list, These Business Services are realized by Technical services.

Include the list of business services impacted with this solution.

Add/Modify /Delete	*Business Service Name*	*Description of the type of change*
Modify	Pega - Real-time decision	Pega enables real-time decisioning will be modified to use the Container API to serve offers to digital channels, Pega will apply business rules, and decision strategies to determine the next-best action for each customer logging into Digital and return the offers to OneApp for presentation.
Modify	*Siebel*	Siebel Interaction history retrieval will be updated to reflect the digital offer interaction along with offer acceptance. The Agents will refer to the interaction history to fulfill customer offers through assisted channels for customer initiated assisted fulfillment.

User Groups Impacted

Required for ARB presentation Next Back

Examples of elements have been listed in the table below. Add, delete or modify the elements to suite the recommended solution.

Roles, functions	Description of the Impacted User Groups
Customer - Domestic Mobile	No Impact

Customer - Domestic Mobile Roaming	No Impact
Customer - International Mobile Roaming	No Impact
Customer - Fixed Consumer	No Impact
Customer - Enterprise & Government	No Impact
Customer - Wholesale Fixed	No Impact
MVNO / MVNE	No Impact
Retail / Dealer	*Update to existing NBA offer fulfillment for digital initiated journeys (training for changes to NBA offer fulfillment)*
Customer Service Rep	*Update to existing NBA offer fulfillment for digital initiated journeys (training for changes to NBA offer fulfillment)*
Customer Online Store	No Impact
Customer Self Care	*Update to digital offer presentation on OneApp (Web & App) to source NBA offers from Pega*
Credit / Collections	No Impact
Business Intelligence	New prepaid data feed to Pega from either the AWS Illuminate platform or Teradata to enable prepaid offers on Digital
Revenue Assurance	No Impact
Deferred Channels	No Impact
Manage Service Partner	No Impact
Workplace	No Impact
Staff / Employees	No Impact
...	*Any other area not covered in the above list*

Roles, functions, and separate table for business interfaces and associated enhancements required

Table 3: User Group Changes

Domains Impacted

Team	Team	Description of the Impact
Network & Infrastructure	Cloud Infra. Engineering (CiNS - CiE)	*Network connectivity between Pega CDH, APIgee and OneApp. Also Pega CDH connectivity with new Data platform (Illuminate - MDS)*
Network & Infrastructure	Ent. Gov. & Wholesale Network Services	*No impact.*
Network & Infrastructure	Fibre Access Networks	*No impact.*
Network & Infrastructure	Fibre Engineering Services	*No impact.*
Network & Infrastructure	Fixed Voice Engineering	*No impact.*
Network & Infrastructure	Mobile Data Engineering	*No impact.*
Network & Infrastructure	Mobile Voice Engineering	*No impact.*
Network & Infrastructure	Network Dev. Engineering (CiNS - NDE)	*Network connectivity between Pega CDH, APIgee and OneApp. Also Pega CDH connectivity with new Data platform (Illuminate - MDS)* Firewall , Private link and raise SSP
Network & Infrastructure	Network Services Engineering (CiNS - NS)	*No impact.*
Network & Infrastructure	OSS Platforms & Design (CiNS - OSS)	*Monitoring of new APIs introduced between Digital (OneApp) & Pega for Offer Presentation & Interaction History*

Network & Infrastructure	Technology Centres & Major Construction	*No impact.*
Network & Infrastructure	Transmission Networks	*No impact.*
Network & Infrastructure	Wireless Networks	*No impact.*
Tech. Service Operations	Network Operations - Fixed	*No impact.*
Tech. Service Operations	Network Operations - Mobile	*No impact.*
Tech. Service Operations	Service Governance	*No impact.*
Tech. Service Operations	Technology Operations IT & Digital	*No impact.*
Tech. Service Operations	Tech. Ops Service Design & Assurance	*No impact.*
Tech. Service Operations	Workplace Health & Safety	*No impact.*
Technology Security	Technology Cyber Security	*Testing for digital changes on OneApp & DEEL APIs (TBC)*
IT & Digital	IT -Sales & Care	*Digital changes to OneApp for Pega offer retrieval* *Digital changes to OneApp for Pega offer Presentation* *Digital changes to OneApp for Pega offer Interaction* *Apigee changes for new digital integration with Pega* *DEEL API changes for Digital Pega integration*
IT & Digital	IT - ERP	*No impact.*
IT & Digital	IT - QA and Testing	*Testing support required for SIT & UAT and release testing (CVT/BVT)*

		Additional testing support for Performance testing & Penetration testing maybe required (TBC)
IT & Digital	IT - Revenue Charging & Integration	*No impact.*
IT & Digital	IT - Workplace	*No impact.*
IT & Digital	IT- Data (Business Intelligence)	*Changes to support prepaid data sourcing on Pega from Illuminate / Teradata*
Finance	Margin Assurance& Fraud Detection	*No impact.*
Customer Ops	Consumer and COPS	*The Consumer CLM team will require update to processes for digital initiated offer fulfillment on Assisted journeys (on Siebel) including referring to Interaction history to refer to digitally accepted offer.* *COPS team will need to referesh training of resources to include fulfillment of digitally accepted offers, and ensure capacity for any campaigns that may introduce additional volume of calls due to digitally presented campaigns.*
Customer Ops	Billing & Payment	*No impact.*
Customer Ops	Business Configuration	*No impact.*
Customer Ops	Care Platform Operations	*No impact.*

Customer Ops	Credit & Collections	*No impact.*
Customer Ops	Customer Care	*Updated training of resources to include fulfillment of digitally accepted offers on assisted channels (Siebel) to include referring to Interaction History to lookup digitally accepted offers.*
Customer Ops	Go To Market (GTM)	*Resources to support GTM strategy for Ignite R3.0* including Training & Content setup on AEM for digital offers (TBC)
Customer Ops	Logistics & Inventory	*No impact.*
Other		

Table 4 Impacted Domains

Process Impacts

Required for ARB presentation [Next](#) [Back](#)

Identify any new processes to be developed or existing processes that will need to be changed. Describe the new processes or changes required to existing processes

ETE process	New / Modified	Description of the change
Digital Offer setup & deployment	Modified	CLM team to configure content on AEM for digital offer presentation to align content presentation as per Pega offers setup, CLM team to engage digital producers to setup the content on AEM CLM team to define CTAs for each digital offer to be presented on OneApp

		CLM team to define CTA action redirection URLs. This can include existing URLs on OneApp, or new content page to direct users to assisted journeys. For New content CLM team to support static content page setup on AEM (TBC) CLM team to direct offer presentation rules on Pega for account level offers containing variety of prepaid and postpaid services (Pega will return the offer prioritization to digital)
Digital Offer retrieval (OneApp ←→ Pega)	Modified	OneApp to decommission RTDM offer retrieval OneApp to replace RTDM service level offer presentation rules with Pega Account level offer presentation rules OneApp to source NBA offer from Pega and display on Dashboard carousel & Offers for you page
Prepaid data feed to Pega	New	Pega will ingest prepaid data sourced from Illuminate / Teradata to allow prepaid offers presentation on Digital
Digital Interaction History Sync (OneApp → Pega)	Modified	OneApp to feed all offer interactions on digital (Dashboard and Offers for you) to Pega (including Displayed, Viewed, Postive - Accepted & Negative - Rejected)
Interaction History Sync (Pega → Siebel)	Modified	Siebel interaction history retrieval and display on Siebel will be updated to inlcude new attributes define channel value and Notes
Assisted NBAA Offer Fulfillment	Modified	Assisted offer fulfillment on Siebel will be updated to includfe new use case for digitally accepted offers as follows: • For all digitally accepted offers that cannot be fulfilled on digital as part of Ignite 3.0,

		end customers will be instructed to contact assisted channels for offer fulfillment • When contacted assisted channel agents will lookup customer on Siebel and retrieve interaction history • Agents will look up digitally accepted offer through Pega Interaction history • Agents will follow the existing Pega Offer retrieval steps to capture Intent and retrieve NBA offers from Pega • Pega presents same Offers to Siebel as presented on Digital (including Accepted offer) • Agents will accept the same offer as already accepted on Digital (referring to the interaction history) • Agents will proceed with rest of the steps for offer fulfillment through Siebel

Table 5: Process Impacts

Critical Success Factors

Required for ARB presentation Next Back

Describe what measures will be used to quantify this initiative was successfully implemented and are meeting associated business requirements.

End to End Solution Description

Required for both SO and SA

Solution Overview

Required for ARB presentation Next Back

The solution replaces the existing RTDM-driven offer management within the MyVodafone (OneApp Web + App) Dashboard experience with a Pega-based decisioning engine. Pega acts as the primary decisioning engine, owning all eligibility evaluation, prioritization, suppression, and offer selection logic. Current design is only covering limited use cases of offer presentation on OneApp only, and will be extended to cover a wider scope in Ignite release 3.1. For this scope, the Digital OneApp offers are strictly presentational, they render offers provided by Pega and perform no decisioning or arbitration. No changes to the digital offer fulfillment is part of the this scope.

Decisioning Flow & Responsibilities: When a customer accesses the Dashboard, the digital channel requests offer decisions from Pega, passing necessary context (customer account identifiers, logged in role, placement details, etc.). Pega evaluates eligibility, applies business rules and adaptive models to prioritize offers, and returns a clear outcome across all services for the customer. If no offers are eligible or Pega does not respond within agreed performance timeframes, the digital channel invokes a defined fallback experience to maintain seamless customer interaction.

Key solution enhancements include:

- Pega Enhancements: Pega will expose the Container API for digital consumption to present account wide offer for all services. this will include offers across both prepaid and postpaid services (minimal prepaid data ingestion via Teradata / MDS to support this). Pega will ingest the digital interaction & impressions data to register customer interactions across presented offers. Offer configuration (eligibility rules, prioritization, suppression logic) is managed entirely within Pega.
- Digital Enhancements:
  - Digital OneApp will update NBA offer retrieval to Pega, however the overall offer retrieval pattern and UX presentation templates will remain unchanged for Ignite 3.0.
  - DEEL APIs for RTDM offer retrieval will be replaced by Pega based API which will work on Account level offers rather than service specifiic offer.
  - OneApp & DEEL APIs will be updated to introduce Interaction & Impression history feed to Pega
  - Apigee APIs will be used as approved integration pattern for all Digital ←→ Pega interactions
  - All offer Content (images) will continue to be hosted in AEM.
- MDS (Illuminate) will ingest a new feed from Teradata for minimal Prepaid attributes to provide to Pega

- The solution enables the CLM business team and relevant operations teams to configure, build, remove, and prioritize offers without introducing logic into digital channels.

Pega Enhancements

- Pega will expose the Container API for digital consumption to present account-wide offers across all services, including both prepaid and postpaid offerings (supported by minimal prepaid data ingestion via Teradata / MDS).
- Pega will ingest digital interaction and impression data to register customer interactions across all presented offers.
- All offer configuration, including eligibility rules, prioritization, and suppression logic is managed entirely within Pega.
- Pega will ingest a minimal set of prepaid attributes via a new feed from MDS (Illuminate), sourced from Teradata, to enable digital presentation of prepaid offers..

Digital Enhancements

- Digital OneApp will update NBA offer retrieval to Pega, while the overall offer retrieval pattern and UX presentation templates remain unchanged for Ignite 3.0.
- DEEL APIs for RTDM offer retrieval will be replaced by a Pega-based API that operates on account-level offers rather than service-specific offers.
- OneApp and DEEL APIs will be updated to introduce interaction and impression history feeds to Pega.
- Apigee APIs will serve as the approved integration pattern for all digital-to-Pega interactions.
- All offer content (images) will continue to be hosted in AEM.

Business Enablement

- The solution empowers CLM business teams and relevant operations teams (Digital & IT) to configure, build, remove, and prioritize offers without introducing any logic into digital channels.

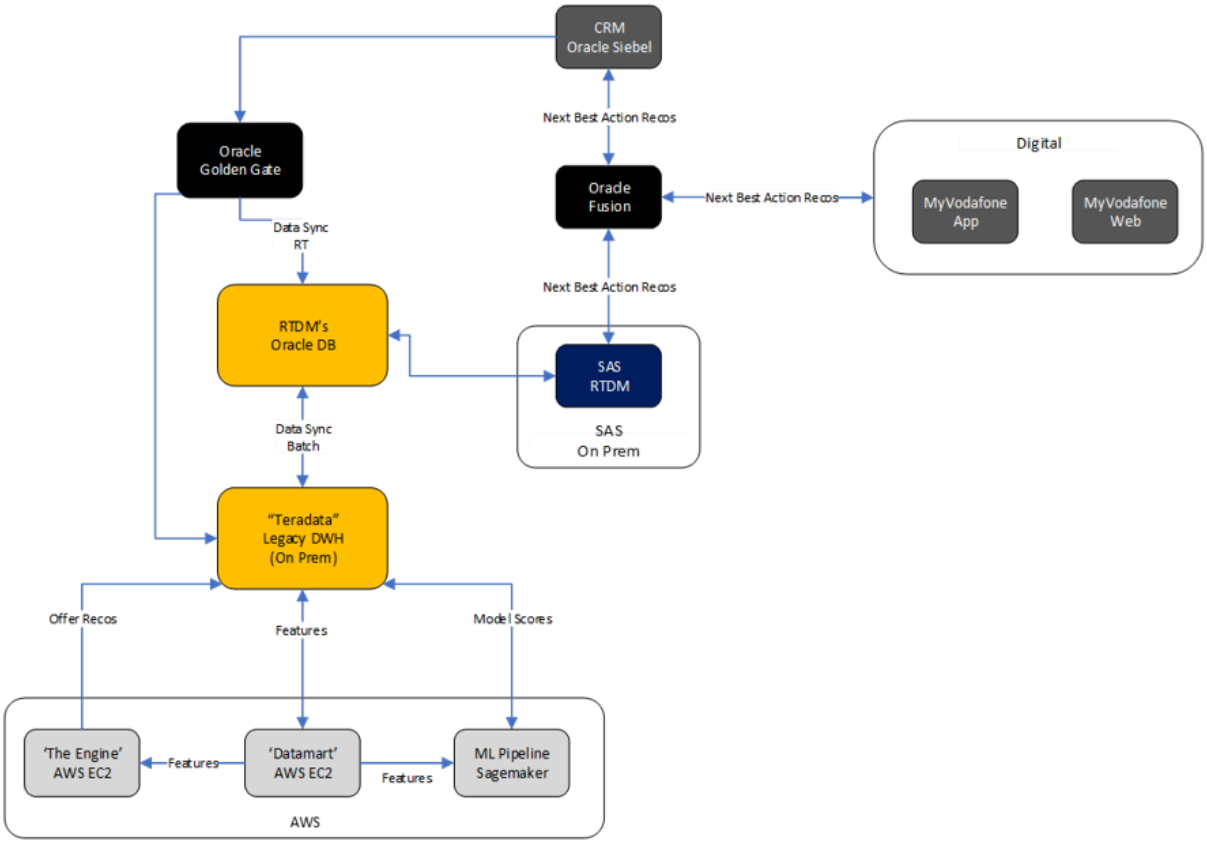


Figure 3: Baseline Solution Architecture Diagram (As is)

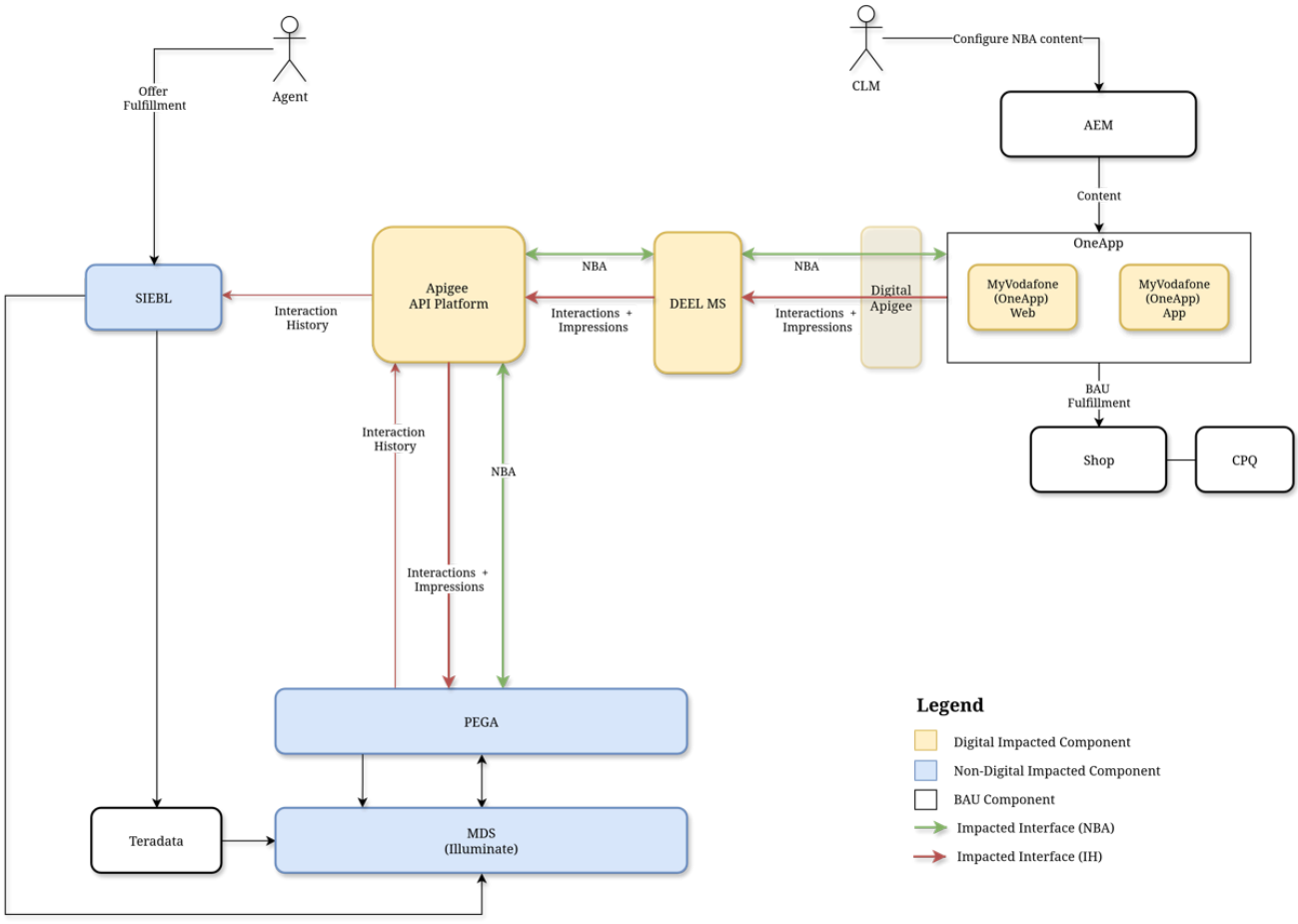


Figure 4: Proposed Solution Architecture Diagram

Alignment to Transition/Target Architecture Blueprint

This solution is classified as tactical. Due to current resource constraints, digital scope has been intentionally minimized. The primary objectives are to:

- Remove the dependency on RTDM for digital offer presentation.
- Deliver a seamless customer experience by ensuring the same offers are presented across all interaction points (assisted and digital).

While the solution aligns with the long-term target architecture, it represents an incremental transition step rather than a full strategic implementation. Ignite release R3.1 aims to provide a fully integrated digital solution with automated digital fulfillment across all interaction points (Shop, Express Upgrade, Vodafone.com.au, etc.). Aim is to reuse the baseline integration built as part of Ignite 3.0 for all future releases, however the approach for digital integration for R3.1 is yet to be finalized.

Solution Assumptions

Required for ARB presentation Next
Back

Where known, provide the solution assumptions for the recommended option.

Assumption	Description	Validation Approach
Fulfillment changes are out of scope for R3.0	Digital fulfillment is out of scope. Anything presented will either be an ATL offer which is digitally fulfillable OR anything that is pega specific will have a CTA to contact live chat, call the contact centre, or visit the nearest retail store to redeem.	Confirmed with Business, and is reflected in the requirements

No Zero rating Impact	Introducing Pega offer retrieval & Interaction APIs calls from digital will not required any impact to existing Zero rating setup (Pega endpoints does not required to be added to Zero rating setup)	Confirmation with digital SME (Rajani), needs to be revisited during detailed design with Digital to further confirm
No CPQ impact	Offer presentation on digital OneApp does not require product setup on digital product catalogue (CPQ). CPQ offer definition will be part of Ignite 3.1 (Drop 2) for Digital Integration.	Confirmed with Digital (Rajani) for current state
No Assisted Fulfillment for Fallback content	Fallback content will not have any manual fulfillment steps, and no interaction details will be sent to Pega for the same	Awaiting Business confirmation
No offer type specific presentation or fulfillment is required	The fulfillment process on Siebel is assumed to be consistent for all offers, irrespective of the offer type and how the offer was presented or accepted in digital. No offer-type-specific fulfillment logic is used in during Siebel offer fulfillment (other than promocode).	Awaiting confirmation from TSI (Raghu / Rakesh)
Offer Fulfillability Risk Due to Missing (or stale) SQ & Coverage Check data	At the time of Pega decisioning, coverage check or service qualification (SQ) data may not be available, or Pega may hold stale data. This creates a risk that an offer deemed eligible during	Acknowledged by Business as limitation. Further assessment is required to determine if any improvements can be made to further optimize the process or handle fulfillment failures gracefully.

	decisioning may not be fulfillable during fulfillment.	
Soft Bundle Fulfillment on Siebel	Soft bundle fulfillment on Siebel requires further assessment, as Siebel processes may need adaptation to support independent fulfillment of loosely coupled bundle components. Discussion with Siebel stakeholders required.	Business Confirmed that Soft Bundles are likely not to be included as part of Ignite 3.0. Further assessment with Siebel SME for the soft bundle fulfillment is required
Offers refresh conditions to be finalized	Detailed list of significant events that should trigger offer refresh on Pega to be finalized as part of detailed design	
OneApp should be able to track significant action completion based on existing functionality	OneApp should trigger offer refresh in case of significant action completion, it is assumed that OneApp can distinguish significant action completion from mid-journey abandoned tasks: OneApp should refresh offers upon completion of a significant action, assuming it can clearly distinguish completed actions (e.g. bill payment, plan change, complaint submission) from incomplete or abandoned journeys. Actions can include: • Lodging a complaint • Submitting a service request • Paying a bill	

	• Resolving an overdue balance • Changing a service or plan state • Offer fulfilment (Completion cannot be tracked on shop is not part of 3.0 scope)	
Unauthenticated offers & dashboard remains unchanged	Currently OneApp (producer?) configures the static offer that will be shown to unauthenticated users. These offers are locally configured in digital and not fetched from RTDM, assuming that for Ignite 3.0 Digital offer presentation, this behaviour remains unchanged.	Confirmed with CLM Team (Poppy), BAU behavior for Ignite 3.0, no change. To be revisited as part of Ignite 3.1, to change to Pega driven offers based on CDP identity stitching data for unauthenticated scenarios.

Table 10: Solution Assumptions

Solution Constraints

Where known, provide the solution constraints for the recommended option.

Constraint	Description	Validation Approach
Minimal UX Impacts on digital	Due to digital resources contraints, the solution is aiming to reduce digital UX changes	Proposed solution for Pega offer presentation on OneApp follows existing UX patterns set by current RTDM offer presentation. Figma designs for OneApp dashboard and offers for you are kept as-is.
No new content type will be hosted on AEM	Current supported content type for AEM self hosted:	

	pdf | jpg | png | svg | webp | doc | docx | zip	

Table 11: Solution Constraints

Solution Pre-Requisites and Dependencies

Where known, describe any pre-requisites associated with the implementation of the recommended solution and the potential impact to the solution if the pre-requisite is not met.

Pre- Requisites	Impact
Prepaid Data availability in Pega	Pega will ingest a minimum set of Prepaid customer data (aligned to current RTDM usage) into Pega (while keeping Prepaid decisioning fully separate from the Postpay CIC model) to present prepaid offers on digital channels. Proposed Approach: SO: Pega CDH Prepaid Data Ingestion Proposed Attributes: Prepaid data required for RTDM replacement through Pega

Table 12: Solution Pre- Requisites

Where known, describe any dependencies associated with the implementation of the recommended solution and the potential impact if the solution is not implemented according to the proposed architecture.

Dependencies	Impact

Table 13: Solution Dependencies

System Register and CMDB

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Required for both SO and SA

Use the below information to classify the current solution SBBs in the system register updates table:

Attribute	How to populate this field?	Acceptable Values
Service Criticality	Derived from BIL questionnaire	Mission Critical Business Critical Standard Support
SOCI Critical System	**For Existing systems**, this can be referred to from system register. **For new systems**, use the tool TPG SOCI critical assets test .xlsx to identify if your system falls under SOCI Critical asset.	Yes No
Data Sensitivity	Based on definition from system register.	Financial Sensitive Network Sensitive PII Sensitive NO Sensitive Data
Data Classification	Based on Information Classification and Handling Policy Section 3B)	whats the Public Confidential Highly Sensitive
Hosting Platform	Based on Solution	Ex: SaaS, PaaS, On Prem, AppCloud, Public Cloud etc..,
System Register Update Required?	Based on Solution	Yes (or) No

Service Criticality	

SOCI Critical System (Yes/No)	
Data Sensitivity	
Data Classification	
Hosting Type	
System Register Updates required?	

System Register Updates

System Register Update

Required for ARB presentation

*Every project is required to ensure that the **existing systems** in scope of their project need to have accurate data in the system register and CMDB. Consequently, the below two are the preconditions for ARB approval of this solution:*

1. ***Systems Register Accuracy***

All existing systems impacted by the solution must have correct and up-to-date accurate data entries in the Systems Register. This includes metadata, ownership, lifecycle status, and other mandatory attributes.

2. ***Workload and resource (subnet, hosts etc..,.) Mapping***

All existing systems within the scope of the solution must have their workloads and hosts created and related to theirsystems in the CMDB. This is essential for infrastructure visibility and effective risk management.

The process for managing the CMDB data along with the respective SSPs is described here. Helix CMDB Data Management Process Guide

	Total # Of Systems	*Total # Of Workloads*
New	*0*	
Existing	*10*	*Pega Customer Decision Hub, Siebel, MyVodafone, My Vodafone Android app, My Vodafone iOS app,* Digital Microservices, *Apigee API Platform, Customer Data*

		Store, Digital Apigee API platform, SAS RTDM
Retiring	*0*	

Confirm the followings:	***Yes***	***No***
I confirm all system mandatory attributes have been revised and updated	*Yes*	
I confirm all workloads mandatory attributes have been revised and updated	*In Progress*	
I confirm all system and workload attributes are accurate and up-to-date.	*Yes*	

The dashboard to check compliance for the systems is available here.

NOTE:

1. *Every project needs to ensure that the above two preconditions are closed for full ARB approval.*
2. *For **New Systems,** this data will be available with the project/architect and it will be captured as part of SSP request to create system/workload. The new system should be in the planning stage. Upon ARB Approval the system shoould be moved to Design stage. This data will be available with the project/architect and it will be captured as part of SSP request to create system/workload.*

Component Overview

This section should capture all the solution building blocks impacted with this solution.

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The following components perform key roles in the delivery of the end to end solution. For Solution Outline it's a high-level overview and should not be taken as an exhaustive list of

impacted components or the impacts on them. For Solution architecture however, a full list of all impacted components along with the impact should be included.

The proposed change to each Solution Building Block (SYSTEM) is categorized as either New, Modified, Decommissioned or Unchanged.

Where relevant the solution components must incorporate Service Security, Service Management and Legal and Regulatory requirements.

Use names as per the system register. Ensure that the solution follows the Architecture Convention Standard.

Application Objects (Application Systems, EDM Workloads)

System(s)

Add/Modify/Delete	System ID Example: SYS000000000309	System Name	System Suite (If Applicable)	Contribution / Impact description including realised capability
Modify	SYS000000001939	Pega Customer Decision Hub		**Decisioning** • Serve as the core NBA engine for digital self-care channel on OneApp (web & App). • Enhanced decisioning strategy to support new digital use cases **Offer Presentation** • Pega operates in a headless mode, to provide digital offers via APIs. • Pega decisioning to provide dynamic Call-To-Action (CTA) labels for offers and recommendations surfaced in OneApp

				• Pega decisioning to provide dynamic CTA destinations, and CTA availability at runtime based on customer context, user permissions, and action intent. **Decisioning changes** • Pega will provide decisioning for product, service, and engagement recommendations at account level for presentation for the digital OneApp. • API payloads from the digital channel must include user role and user type attributes. • Decisioning strategies will be updated to support three user roles with different permission profiles (Account Holders, Authorised Users and Limited Users) • CTA availability becomes dynamically controlled by Pega via business rules, not hardcoded in digital channels. • Eligibility rules must evaluate account attributes combined with user role permissions. • Interaction history must be enriched with user role (TBC). • No CTA may be enabled for limited users unless explicitly overridden by Pega business rules.

- Offer presentation for limited users is read-only; no action or fulfilment initiation is permitted via digital.
- Pega decisioning for limited users is restricted to informational or engagement recommendations only. (TBC)

API Integrations

- New APIs exposed to Digital (OneApp) via APIgee:
  - `Container API` – fetch NBA recommendations
  - `CaptureResponse API` – capture Interaction events including offer responses (accept/reject) and impressions (displayed/viewed)
- Existing Pega Interaction History (IH) API will be updated to include additional Channel & Notes detais to Assisted (Siebel)
- Supports real-time interaction with Digital OneApp.

Cloud & Environment Changes

- Support updates to Pega cloud and interfaces with existing and new systems.

Data Jobs

- Ingest the minimum set of Prepaid customer data (aligned to current RTDM usage) into Pega, while

				keeping Prepaid decisioning fully separate from the Postpay CIC model. • Decisioning for Prepaid will be rule-based and isolated to avoid impacting the existing CIC design. • Minimal set of attributes for Prepaid: Prepaid data required for RTDM replacement through Pega (TBC)
Modify	SYS2285	Customer Data Store (Illuminate)		• Provide the minimum set of Prepaid customer data (aligned to current RTDM usage) into Pega (placeholder, to be detailed in the Data SAD) (TBC)
Modify	SYS0000000001207	Apigee API Platform		• Enable proxy APIs on Apigee to allow digital OneApp (MyVF) to consume the following Pega APIs: ◦ **Container API** – for NBA recommendations ◦ **CaptureResponse API** – for Interaction & impression data flow such as accepted/rejected offers, displayed , viewed offers • Update existing proxy APIs on Apigee for payload changes to include new paramters for Pega Interaction historyt (Customer offer history API) • Includes configuration of API security policies, and API

				lifecycle support (if applicable). • Facilitates API gateway services for Pega-OneApp integration.
Modify	SYS000000000387	MyVodafone (OneApp)		Self-Service (OneApp) is the target state self-service web and native mobile app platform for all three brands on the consumer stack (Vodafone Mainstream, DFA and 3rd brand). **Offers Retrieval & Presentation** • Offer retrieval via DEEL MS to Pega (Apigee), passing account ID and user role; re-queried on account switcher • Top 4 ranked offers from Pega displayed as dashboard banner carousel • "See All" view displays up to 20 offers ranked by Pega priority • "View All" link shown only when >3 offers exist • Offer heading and background image URL from Pega; images sourced from AEM • View all will be displayed only if there are more than 3 offer cards and will not appear if there are 3 or less. • Further details covered below: SAD - Ignite R3.0 Di

gital Offer Presentation | One App Offers Rerieval & Presentation

Offer Notification Rule

The existing Offer Notification rule will remain unchanged and continue to operate based on current notification logic as follows:

- A message icon is displayed on the menu when the customer has one or more eligible offers.
- The icon is visible only when the customer has unread offers.
- The number of unread offers available for customer viewing is determined by Pega CDH (replacing SAS RTDM).
- The icon is removed upon when all offers are read.
- When the customer has multiple unread offers, the icon displays the count of unread offers.
- The offer count presented in the icon updates dynamically as the customer views each offer.

Access to Offers for different user roles

- Role will now be handled in Pega and OneApp should

				only enforce the rules directed by Pega in the resposne payload. • Pega will drive the CTA action eligibility across offers fo all user roles - Account Holder, Authorised Contact, Limited User • API payloads from the OneApp to Pega must include user role and user type attributes. • Pega will consider the user roles with different permission profiles (Account Holders, Authorised Users and Limited Users) and define CTA availability dynamically via business rules, not hardcoded in digital channels. **CTA Configuration & Interactions** • Dynamic CTAs: OneApp renders CTA label, destination, and availability dynamically from Pega payload, no hardcoded CTA logic or role-based rules in OneApp • CTA Routing: OneApp should support positive CTA destinations (website, live chat, call, shop journeys) and negative CTA labels as defined by Pega

- Event Transmission: OneApp to Send Positive (accept), Negative (dismiss), and Neutral (tile click) events to Pega in real time
- UI Updates: Removes accepted/dismissed offers from view and updates count upon successful transmission to Pega
- Impression Capture: Sends real-time impressions to Pega only when offer tiles are physically visible on screen (carousel load, scroll, expanded view); no off-screen impressions
- De-duplication: Prevents duplicate impression events for the same offer within the same session
- Further details covered below: SAD - Ignite R3.0 Digital Offer Presentation | One App CTA Actions & Behavior:

Fallback Offers

- Digital Fallback offers to be mantained as part of OneApp as static offer content to be displayed only when Pega response is empty (by design), times out, exceeds SLA, or technical error occurs
- Pega content (offers or configured fallback) always

takes precedence over digital fallback

- Further details covered below: SAD - Ignite R3.0 Digital Offer Presentation | Fallback Offer Handling

Refresh Offers

- OneApp should track and classify events as significant events, which can include substantial event such as accept / dismiss and offer, fulfilment journey completion, live assisted interaction end, or service/account change, bill payment, lodging a complaint. (Detailed list of significant events to be finalized as part of detailed design).
- OneApp should trigger re-decisioning request to Pega when customer returns to dashboard or navigates back to offers section, after one or more significant actions have been completed
- OneApp does NOT trigger re-decisioning for: scrolling, impressions only, expanding/collapsing tiles, viewing informational content, passive navigation, or mid-journey steps
- Upon session expiry and new login, OneApp re-requests

				offers from Pega (BAU behaviour) **Toggle Capability for OneApp changes** • OneApp to support toggle flag to control decisioning source (SAS RTDM vs. Pega) for the above changes • This is required to independent digital releases (Pega path deployed but disabled) and rapid rollback without redeployment occur with Pega integration, without requiring a digital release Offer types and details covered below: SAD - Ignite R3.0 Digital Offer Presentation | OneApp Offer Categories (Presentation)
Modify	SYS0000000000389	My Vodafone Android app		Platform specific build for the My Vodafone Android App for incorporating the Offer presentation changes covered for MyVodafone
Modify	SYS0000000000390	My Vodafone iOS app		Platform specific build for the My Vodafone iOS App for incorporating the Offer presenation changes covered for MyVodafone
Modify	SYS2032	Digital Apigee API Platform		APIGEE, TPG's strategic API management platform will be used as the API management platform for OneApp APIs. Apigee Hybrid runtime plane is deployed on one of the TPG's

				AWS Account on EKS inside a VPC. • New proxies for the Pega interaction & offer APIs
Modify	SYS2022	Digital Microservices (DEEL MS)		DEEL-MS layer includes APIs and Integration services required for OneApp to support self-service function. • DEEL will not use cache to cache Pega responses • DEEL APIs layer will be introduced for all calls between OneApp & Pega for Offer retrieval and Interaction udpates
BAU	SYS2025	Digital Adobe Experience Manager		AEM is Central Content Management system that acts as single source of content for SHOP, ONE APP and any other system that would need the content. This will continue to be used for the offer presentation on OneApp, following the existing pattern in place for RTDM offer presetnation. Pega will provide the offer details along with AEM hosted URL for offer content to OneApp. For OneApp (BAU) • The headless content will be managed as content fragments and the OneApp will consume the headless contents via GraphQL endpoints

				No System change, only content change based on new offer presentation (BAU).
Modify	SYS000000000454	Siebel		**Interaction History** • Update to the existing past Pega interactions via Customer Offer History API to include 2 new parameters (Channel Id & Notes) • Update to the Display in Customer Overview and CA UI tabs to assist agent context to include the new parameters (Channel Id & Notes)
Modify	SYS000000001617	SAS RTDM - Real Time Decision Management		The system supports the execution of legacy campaigns and facilitates real-time decision-making across various channels, apart from Siebel (Agent-Assisted). This change will remove SAS RTDM flow from OneApp. • Decommission SAS RTDM API call to OneApp (MyVF)

EDM Workload(s)

Add/Modify/Delete	Workload ID Example: WLD1952	Workload Name	System ID this workload is related to	Target Hosting Platform (Ex: TPG AWS, GC,	Contribution / Impact description including

				VVF etc..,.)	realised capability
Modify	SYS000000001939	Pega	The workload is associated with the PEGA Decisioning platform, which consumes customer data from upstream systems to enable real-time decisioning and personalised outcomes. PEGA interacts with OneApp, Siebel, Braze, and Illuminate to evaluate customer data and orchestrate decision-driven actions across channels.		Pega system consists of CDH (Customer Data Hub) which does the decisioning, API endpoints and Data ingestion from data pipeline. Pega Container APIs will be used to pressent the offers on Digital
Modify	SYS2032	Digital Apigee API Platform	APIGEE, TPG's strategic API management platform will be used as the API management platform for OneApp APIs. Apigee Hybrid runtime plane is deployed on		

			one of the TPG's AWS Account on EKS inside a VPC.		
Modify	SYS00000000 1207	Apigee API Platform	APIGEE, TPG's strategic API management platform will be used as the API management platform for all downstream interactions with Pega across multiple interfaces		

Technology Objects (Network System)

Network System(s):

ADD/MODIFY/DELETE	Network System ID *(Ex: SYS2175)*	Network System Name *(Ex: GC60 Core Switch)*	Type of Network System *(Ex: Core Switch Nodes)*	Comments
		Please use the same object name as defined in the system register		

Data Assets

Name	Proposed Change	Sensitivity	Related System(s	Contribution /	Physical location	Who has Access	Privileged/Admin

			)	Impact Description	where DATA is stored (data sovereignty)	Access Controls?	Access path
e.g. CDR	Format Change	PII PCI Network Corporate Other	Dataware house	New field added Changes to access Change to grant of access	AWS Australia, TPG on-prem DC	TCS, TPG internal support staff Remote access via Citrix Authentication via AD	Access via CyberArk Or Access Via Standard Windows/ Linux Jump Host Access Via Netskope

Table 15: Data Component Contribution / Impact

Solution Interactions

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Table 14: Active Component Contribution / Impact

New or Proposed Change	Interaction Method (Ex: API, HTTPS, MFT etc..,.)	From Workload	To Workload	Relationship type (Ex: Data Sharing, Orchestration etc..,.)	Provider (which workload)	Contribution / Impact Description including impacted realised capability	Version
New	API	MyVodafone (OneApp)	Digital Apigee API Platform	NBA Offer Retrieval	Digital Microservices (DEEL MS)	Interface for Offer retrieval from Digital (OneApp) to DEEL MS via Apigee. DEEL MS will provide the abstraction and translation required for the this API for interfacing with Pega	
		Digital Apigee API Platform	Digital Microservices (DEEL MS)				

New	API	MyVodafone (OneApp)	Digital Apigee API Platform	Customer Interactions	Digital Microservices (DEEL MS)	Customer interaction Interface for providing customer interactions and impressions data from digital (OneApp) to DEEL MS via Apigee. DEEL MS will provide the abstraction and translation required for the this API for interfacing with Pega	
		Digital Apigee API Platform	Digital Microservices (DEEL MS)				
New	API	Digital Microservices (DEEL MS)	Apigee API Platform	NBA Offer Retrieval	Pega	Interface for Offer retrieval from Digital to	

		Apigee API Platform	Pega			Pega, via Apigee.	
New	API	Digital Microservices (DEELMS)	Apigee API Platform	Customer Interactions	Pega	Customer interaction Interface for providing customer interactions and impressions data from Digital to Pega via Apigee.	
		Apigee API Platform	Pega				
Change	API	Siebel	Apigee API Platform	Customer Offer History API	Pega	Provide visibility into past interactions. This API will be updated to include new parameters required for Digital Interactions	
		Apigee API Platform	Pega				

Component Interactions

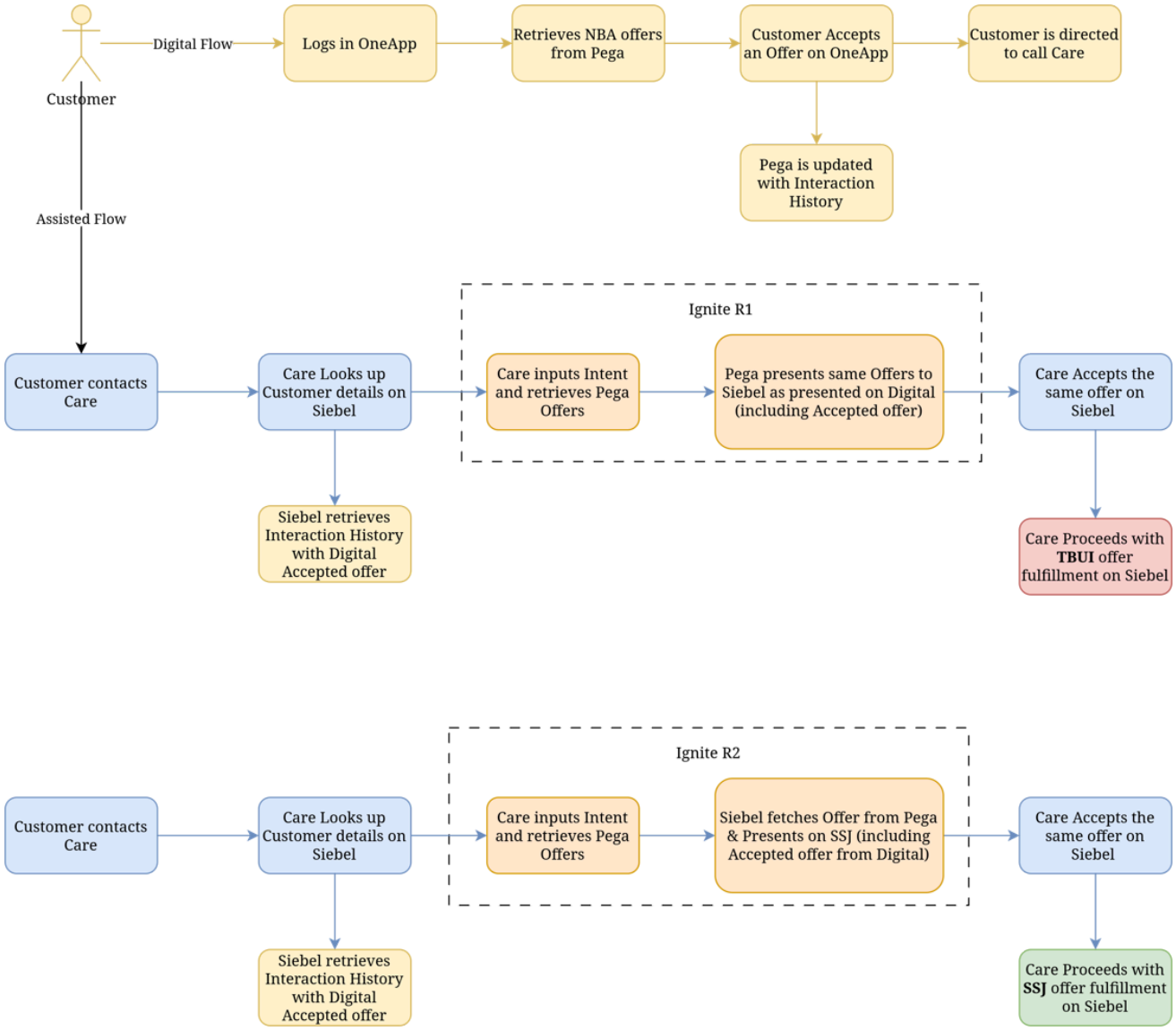
Service View – Describe how the different components will interact with each other to realise the business capability. Alternatively, please include the service view in the Solution architecture diagram. Ex: Sequence Diagrams, Service Flow diagrams etc..,.

Digital Offer Fulfillment Process:

The introduction of digitally accepted offers that cannot be fully fulfilled within the Ignite 3.0 digital experience will impact the assisted offer fulfillment process in Siebel as follows:

When a digitally accepted offer cannot be fulfilled digitally, the following process changes apply:

1. OneApp Postive CTA: When end customers accept an offer on digital, the CTA will allow to call assisted (Call Assisted embeded in ETA) or redirected to static instructional content to contact assisted channels to complete fulfillment of their accepted offer.
2. Agent lookup on Siebel: Upon customer contact, agents will retrieve the customer record in Siebel and review interaction history to identify the previously accepted digital offer.
3. Offer retrieval from Pega: Agents will follow the existing steps introduced by either Ignite R1 (embeded Pega screesn) or Ignite R2 (Pega headless mode) for Pega offer retrieval steps to capture customer Intent and retrieve NBA offers from Pega.
4. Offer presentation: Pega will present the same set of offers to Siebel as were presented on digital, including the previously accepted offer.
5. Offer acceptance: Agents will accept the same offer already accepted by the customer on digital, referencing the interaction history for verification.
6. Fulfillment: Agents will proceed with the remaining offer fulfillment steps within Siebel as per existing procedures (for Ignite R1 or Ignite R2).



Key Impacts to Roles and Systems

Impact Area	Description
Agent (Siebel)	No new interface required; agents use existing Pega integration and Interaction History lookup. Only change is the new attributes added to the interaction history to identify digital interactions.
Agent workflow	Additional step added to verify digitally accepted offer via interaction history before proceeding with standard fulfillment.

Pega	Must retain and expose digitally accepted offer status within Interaction History for retrieval by Siebel.
Digital experience	Customer communication (instructions to contact assisted channels) must be added for applicable offers.
Training	Agents may require awareness training on the new hybrid fulfillment scenario and how to locate digitally accepted offers in Pega Interaction History.

Known Issues:

- Having process driven assisted fulfillment for digitally presented offers may result in poor customer experience
- Agents lookup of digitally accepted offers will have to rely on the Interaction history lookup and manual identification of the accepted offer, the Pega retieved offers will not have any visual cues to distinguish the accepted offers
- Agents acceptance of the same offer is a manual step, and can cause issues due to human errors of accepting a wrong offer (e.g. in case of similar offers)
- Duplicate offer acceptance events will be sent to Pega (digital + assisted) for the same offer
- Even when an offer is accepted digitally, Pega will only reflect fulfillment via assisted journeys

OneApp Offers Rerieval & Presentation

- **Offer Retrieval & Account Context**
  - When a customer logs into OneApp, DEEL MS queries offers for the customer account and passes the customer account ID and logged-in user role details to Pega.
  - When the user switches accounts using the account switcher, DEEL MS queries offers from Pega for the newly selected customer account.
  - Pega returns offers for all services under the customer account via Apigee.
- **Dashboard Presentation (Banner Carousel)**
  - The top 4 offers received from Pega are presented on the MyVodafone dashboard as a digital offer banner carousel.
  - Pega ranks offers across the customer's entire account context (services, eligibility, interactions, suppression rules).
  - When the customer clicks a banner, they are taken to the offers page with the accordion open for the selected offer.

- **"See All" Offers View**
  - OneApp displays a ranked list of up to 20 offers in the "See All" view.
  - Offers are ordered based on the priority ranking received from Pega.
  - Unread offers are displayed first, in priority order returned from Pega.
  - The "View All" link is displayed only when there are more than 3 offer cards; it is hidden when there are 3 or fewer offers.
- **Offer Content & Rendering**
  - Offer heading is derived from the Pega payload for each offer.
  - Background image URL is provided by Pega, with the image asset sourced from AEM.
  - Customers can expand an accordion to view further details associated with the offer template based on service type (e.g., postpaid, prepaid).
- **Interaction Tracking**
  - For all offer interactions (e.g., banner display, offer view, acceptance, decline), the status is sent to Pega.
  - When a customer accepts or declines an NBA offer from the banner, the banner disappears and the interaction status is recorded.

OneApp CTA Actions & Behavior

- OneApp should render CTAs dynamically based on configuration provided in the Pega decisioning response, including CTA label, destination, and availability
- No CTA logic (label, destination, availability, or role-based permissions) is hardcoded in OneApp; all behaviour is driven by Pega at runtime
- Pega determines CTA availability based on customer context, user role (account holder, authorised user, limited user), and OneApp should enforce the CTA availability based on Pega offer payload (e.g. CTAs disabled or hidden for limited users)
- Positive and negative CTAs labels are defined as per Pega provided payload, and OneApp should apply the CTA Labels based on Pega response for both positive (e.g., "Buy Now", "Chat to Vodafone", "Click to Call") and negative (e.g., "I'm not interested", "No thanks") CTAs
- Postive CTA destination URL should be defined based on Pega provided payload (aligned with RTDM behavior), OneApp should support routing to:
  - Website (deep link or external URL)
  - Live Chat
  - Call (phone number)

    - Shop journeys
- For all interactions with offers, OneApp should trigger an event with action details to Pega (e.g. Postive, Negative & Neutral events should be sent to Pega in realtime)
- Payload includes customer, offer instance, and decision interaction identifiers for accurate linkage
- **Postive Event:**
    - Real-time transmission of positive customer responses (accept/positive CTA) from OneApp to Pega CDH upon selection in the expanded "For You" view
    - Positive CTA treated as explicit acceptance, distinct from neutral signals (e.g. view, dismiss)
    - CTA label, behaviour, and destination are dynamically provided by Pega and rendered by OneApp without interpretation
    - UI updates (e.g., offer removal, count adjustment, list refresh) triggered by OneApp upon successful transmission to Pega
- **Negative Event:**
    - OneApp should send Negative Inteaction event as real-time transmission upon (dismiss/negative CTA) to Pega CDH in the expanded "For You" view
    - Negative CTA treated as explicit negative response, distinct from neutral signals (e.g. offer tile click to expand)
    - Upon successful transmission, OneApp removes the dismissed offer from the current view and updates the remaining offer count
    - Pega uses the negative response for suppression or arbitration logic, OneApp performs no local suppression decisions
- **Neutral Event:**
    - OneApp should send Neutral Interaction event as real-time transmission to Pega CDH when a customer interacts with an offer tile (offer tile click to expand) in the "For You" section (Web & App)
    - Offer tile click treated as an expression of interest, distinct from accept/positive or dismiss/negative responses
- **Impressions Event:**
    - OneApp captures and sends impression events to Pega CDH in real time only when an offer tile is physically visible on screen
    - Impressions are not sent for off-screen content or tiles that fail to render
    - Payload includes location identifier (dashboard carousel, See more, expanded view), position/index metadata, customer ID, offer ID, decision reference, and channel (Web/App)

(details to be covered as part of the interface contract)

- On dashboard carousel load, only the first visible tile generates an impression; scrolling/swiping triggers impressions as additional tiles become visible
- Expanded offer tile view generates a separate impression linked to the originating offer instance
- OneApp should not trigger duplicate / repeat impressions events for the same offer during the same session (De-duplication rules should be enforced to prevent this behavior)
- Location/placement naming conventions will be finalised during detailed design

OneApp Offer Categories (Presentation)

- **Singular Offer**
  - OneApp presents singular offer recommendations from Pega in the "Offers For You" section as individual personalised offers (not part of hard or soft bundles)
  - Singular offers include standalone plans, SIMO plans, device offers, and Type 0–4 offers across all categories
  - OneApp applies no bundle-specific rendering logic; singular offers are displayed using standard offer tile presentation
- **Hard Bundle**
  - OneApp presents hard bundle recommendations from Pega as a single, inseparable offer tile in the "Offers For You" section
  - Hard bundles comprise one or more services/recommendations that must be treated as a single offer; OneApp does not split or separate bundle components
  - All customer interactions (view, click, accept, dismiss) with the bundle tile are attributed to the originating Pega offer (see details in section: SAD - Ignite R3.0 Digital Offer Presentation | OneApp CTA Actions & Behavior)
  - OneApp applies no bundle-specific rendering logic beyond standard offer tile presentation
- **Soft Bundle (TBC)**
  - OneApp presents soft bundle recommendations from Pega as a single offer tile in the "Offers For You" section
  - Soft bundles contain two or more related offers that are independently fulfillable and priceable (customers can take individual offers or the full bundle)
  - OneApp CTAs (positive/negative) apply to all actions within the soft bundle collectively (aligned with Single offer/ Hard Bundle behaviour)

    - All customer interactions with the bundle tile are attributed to the originating Pega action (see details in section: SAD - Ignite R3.0 Digital Offer Presentation | OneApp CTA Actions & Behavior)
    - OneApp applies no bundle-specific rendering logic beyond standard offer tile presentation
- **Service & Engagement (Non-Fulfillable)**
    - OneApp displays service and engagement recommendations (classified as non-fulfillable by Pega) in the "For You" section (Web & App)
    - These recommendations are rendered based on metadata supplied by Pega, intended for engagement, education, or self-service rather than conversion
    - OneApp applies no special handling or behavioural differences for non-fulfillable recommendations; they are treated as standard offer tiles for display purposes
- Following Offer types are part of the scope for Digital 3.0

Type	Desc	In Scope	Digital Fulfillment
Type 0	Base Plan/Product, No Discount (Standard Pricing)	Yes	Yes, per BAU
Type 1	ATL without promocode - Discount automatically applied in cart (hatcat)	Yes	Yes, per BAU
Type 2 (ATL)	ATL with a promocode	Yes	Yes, for offers already in CPQ / Assisted for rest
Type 2 (BTL)	BTL with a promocode	Yes	Yes, for offers already in CPQ / Assisted for rest
Type 3	Pega discount (ABTL)	Yes	No, Assisted Only for 3.0
Type 4	Adjustment Credit (Standalone Rewards from Pega)	Yes	No, Assisted Only for 3.0

Fallback Offer Handling

Digital will need to present Fallback offers in scenarios where Pega offer retrieval is not possible. Below establishes a clear precedence hierarchy for Digital experience when no eligible offers

are available from Pega CDH. Digital fallback content is displayed only when Pega cannot provide a usable response, and is never combined with Pega-provided content.

Order	Condition	Content Source
1	Pega returns offers	Pega CDH only
2	Pega intentionally returns no offers	Digital fallback
3	Pega response times out or exceeds SLA	Digital fallback
4	Technical error or aunprocessable response	Digital fallback

Key Design Decisions

- Mutual exclusivity enforced: Digital fallback and Pega content are never displayed together in the same render state.
- Digital does not supplement: When Pega returns any content (offers or configured fallback), digital adds no additional tiles or fallback content.
- Digital does not alter: Pega responses are rendered exactly as returned, with no removal, replacement, or reordering.
- SLA-defined timeout: Digital fallback is triggered immediately upon SLA expiry, no waiting beyond defined threshold.

Technology Service KPIs

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Describe the changes (if any) to Technology KPIs due to proposed solution and its components

The list of Technology Service KPIs (Network and IT) reported to execs can be accessed here. If a new line of business/function is being introduced and no existing Service KPIs are available, it will be added through Operations Service Management Process.

***NOTE*:** The project has the responsibility for the following:*

1. *Ensure that the Service KPIs reported to execs are available with the proposed solution and clearly articulate any impacts to those KPIs.*

2. *Any new Service KPIs being introduced by the solution need to be captured as part of Non-Functional Requirements and adhere to the ODS (Operational Design Standards).*

S. No	Impacted KPI	Variations (different Scenarios)	Impact (Functional change or Regression test or NA)

Table 17: Impacts of KPIs

Architectural Decisions

List all the architectural decisions taken to address architecturally significant requirements

Solution Scalability

Clearly specify how the solution meets the scalability requirements based on the expected business forecast. Clearly specify any limitations that may impact the solutions from scaling.

Shared Infrastructure

Clearly specify what environment will be used for implementing the solutions. If not implemented on TPG Telecom's Internal Virtual/Cloud environment clearly specify what are key limitations or drivers.

Re-use of existing Architecture Capabilities

Clearly specify how the proposed solution leverages existing capabilities for the proposed solution. State the reasons, limitations and drivers if a duplicate capability is being proposed.

Cost Efficiency

Licensing

Does the solution require any software licenses or will change current software licenses? If yes, describe how the cost for the software licenses has been optimised

Operating Expense (OPEX)

Does the solution require any ongoing opex or will change current Opex? If yes, describe how the Opex has been optimised.

Cost efficient Solution

Describe the cost efficiencies embedded in the solution architecture and design. Have you considered cost optimisation in the design? E.g.:

- *the use of serverless services? E.g., only consume resources for the compute that is required,*

- *the mechanical sympathy of the design, is it the most efficient database type for the service? e.g. Using NoSQL for maintaining state*
- *Reviewed the logging metrics parameters to balance the cost vs data collection parameters*
- *Use of managed service APIs to focus on the value of the service vs maintenance of the underpinning service components*
- *End to end service cost attributed to the business/technology service driving the demand*
- *Schedule the startup/shutdown of resources to align to demand, e.g. schedule development environments to be stood up during development hours and shutdown overnight*

Other Decisions

List any other architectural decisions that played a major role in the way the solutions were designed.

Open Solution Issues / Items

List any open issues still to be resolved at the time of document sign off.

Issue	Priority	Description	Resolution
	High/ Medium/ Low		

Table 18: Open Issues / Items

Architecture Deviations

(Optional for SO required for SA)

Required for ARB presentation

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Description of Architecture Deviation:

Please include the details of how this solution impacts the target state architecture for the domain (Example of Target state programs are Imagine (hyperlink), Target Data Architecture (TDA) hyperlink etc..,)

. The following are the possible answers:

- *Aligned to the target state architecture with no deviations or technical debts.*
- *This solution will result in a CR on target state program.*

- *This solution will be a regret spend as it is already in scope of target state program. (Provide the reason why this solution can't wait for Target State Program).*
- *This solution will introduce a Technical Debt. If Yes, please provide details of the technical debt below.*

Description of the Technical Debt: (*please include architectural debt as well as debt related to implementation approach.)*

Technical Debt Name	Debt Owner	Impact	Architectural Risk

Remediation Plan

Technical Debt Name	Debt Owner	Remediation Plan	Scope	Cost Estimate (optional)	Time Estimate (optional)

Table 19: Technical Debt

Architecture Policies Standards and Guidelines

All the policies, standards and guidelines published are available in Central Policy Hub on intranet.

https://tpgtelecom.sharepoint.com/sites/Policies/

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Policies Policies

Name	Policy Name	Compliant (Y/N)	Comments
TO_POL_01	Technology Resilience Policy		
TO_POL_03	Service Assurance Policy		
TS_POL_33	Cyber Security Policy		

TS_POL_36	Secure Software Development Policy		

Table 20: Architecture Policies

Standards Standards

Name	Policy Name	Compliant (Y/N)	Comments
CINS_STA_03	Inventory Management Standard		
CINS_STA_01	IP Addressing Standard		
CINS_STA_02	Quality of Service Standard		
CINS_STA_03	Workload Placement standard		
CINS_STA_04	Standard Operating Environment (SOE)		
CINS_STA_05	Cryptographic Standards - Technical Implementation		
IT_STA_01	Information Management Standard		
NPE_STA_01	NPE IT Environment Standard		
TSO_STA_01	Operations Design Standards		
CINS_STA_06	Cost Optimisation Standard for Public Cloud Infrastructure		

Table 21: Architecture Standards

Guidelines Guidelines

Name	Policy Name	Compliant (Y/N)	Comments
CINS_GDL_01	Automation Guideline		

CINS_GDL_02	Configuration Management Guideline		
IT_GDL_01	New Capability Guideline		

Table 22: Architecture Guidelines

Proposed Solution Non-Functional Overview (Relevant areas required)

Required for SA

This document describes the Solution Architecture. Hence all relevant non-Functional aspects must be described

Availability

Clearly specify how the solution meets the following availability requirements based on the business criticality of the solutions (please refer to policies APP-ENT-10 and APP-ENT-11):

- *99.99% availability for any application or infrastructure supporting Mission and Business critical services*
- *Solution to be architected with sufficient redundancy such that no single component, group of components or single site outage will impact the provision of services to TPG Telecom customers.*
- *No Single Points of Failure in the Production Environment*
- *Solution to be architect such that any requirements to upgrade or maintain the application can be performed in a modular process with no impact to customer-facing and critical business services support or delivered by the solution*

If the solution hosted in AWS or other Public Cloud provider, the Public Cloud provider SLA's should be clearly specified. If solution leverages a SaaS deployment, the SaaS vendor committed availability should be specified. Clearly specify if the solution does not comply with associated architecture policy and standards.

Archive and Purging

Clearly specify how the solution complies with the TPG TELECOM Data Retention policy, what scheduled tasks should be in place to perform archiving and purging in automated and scheduled manner without impacting performance or availability or end customer

responsiveness. Clearly specify if the solution does not comply with associated architecture policies.

Error Handling

For Business Process or Service added or customised, Distinct Error Code and Error Messages should be generated for every failure scenario, across Application stack and meaningful message displayed on front end for end customer consumption.

Each Error should have unique code and message to be stored in respective originating application for debugging and reprocessing. In Event of high volume of errors, Application should not crash, jeopardising healthy transactions or background processes.

System should have capability to clear Errors, Failures, Retry automatically or with minimal manual intervention once error causing cause is alleviated (e.g. failed downstream component is up or failed file transfer)

Error Conditions, its message and way to reprocess should be documented in SOP and demonstrated to Operations prior to go live.

Database

Ensure, all database optimisation and performance/tuning parameters specified in the TPG TELECOM Oracle Database SOE have been implemented and:

(a) Tested for effect (i.e.) with the parameter set to the Oracle default and with the parameter set to the TPG TELECOM SOE value – if different.

(b) CPU, IOPS and elapsed time recorded using the default and TPG TELECOM-recommended parameters against a known test suite of database transactions.

(c) Document assumptions and the methodology to determine how estimates are calculated for the database concurrent session limit parameter.

(D) Optimization methodology of databases

(E) All database housekeeping activities required to maintain the integrity and performance of the database must be able to run at any time without impact to customer-facing services

Deviation from Standard Solution Support Model

Describe any non-standard support task under each level

Level	Key Activities	Description
Level 1		

Level 2		
Level 3		
Level 4		

Table 23: Solution Support

Operational Support Functions

Describe the different tools that will be utilised to provide the Operation support functions.

		System 1	System 2
Service Monitoring	Target System		
	Alerting		
	KPI Reporting		
Performance Management	Performance Monitoring		
	Alerting		
Capacity Management	App Capacity Reporting		
	Infra Capacity Reporting		
Fault Management	Target System		
	Auto Ticketing		
Restoration	Backup solution		
Security	SIEM Integration		
	Active Threat Management		
	Forensics		

Access Management	User Identity access via SailPoint		

Table 24: Operational Support Functions

Operations Asset Inventory Management

(Mandatory for SO/SAD)

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Based on the inventory management standard published on the intranet, please specify the target inventory system that is impacted with this solution:

The list of inventory management systems is based on the Inventory Management Standard that is published in the Central Policy Hub.

Asset Management Type	Definition/Examples	Target	Impacts on Target system
CMDB	Physical or virtual configurations items (CIs) required to support the operational ITIL processes.	BMC Helix	
Data Centre Infrastructure/Cabling	Database of equipment, racks, suites, cabling and ancillary equipment in supported Data Centres	Cormant-CS or LSD	
Active RAN and Tx Network Inventory	Inventory Database which records the telco active network element inventory	Orion-T / Orion- R / LSD / InfraDB	

	information including rack, subrack, Card location, manufacturer, type, serial number and connectivity		
Passive RAN and Tx Network Inventory	Inventory database which records passive equipment Eg Antennas, wave guides, batteries, huts, pipes, pits	Orion-T / Orion- R / PipeDB / ArcGIS / LSD / InfraDB	
Telco Spares Management	Database of spares location, status, numbers	SPMS (VHA)	
Telco Build Inventory	Database containing material on hand (not yet installed) to build sites eg racks, huts, antennas, cable	CEVA / Orion-R	
Software Licenses	Database that records software license entitlements, usage, license expiry dates, etc	SNOW	
Employee Asset Management (Ex: PCs)	The database of what equipment is held by employees such as PCs	SNOW, BMC Helix, JAMF, InTune	
Leased Lines	Leased lines used by TPGT, cost, lease dates, lessor details	Orion-T (VHA)	
IPAM			

ERP Cloud (Oracle Cloud)			

Asset Inventory Management

Service Dashboards & Alerting

The following to be considered in the proposed solution in case business services or technical services are modified or newly created:

1. *Any newly on-boarded system or change in existing system impacting listed Service transactions needs to be integrated into Splunk providing transactional activity logs & metrics.*
2. ***Monitoring dashboards** should be designed for the monitoring of critical business services. These dashboard should provide a view of the business services status so that service interruption/degradation that are not detect through alerting can be identified by L0 monitoring teams / NOC teams.*
3. ***Troubleshooting dashboards** should be designed for the purpose of providing a view of technical services that support business services and assist with identifying the root cause of an incident.*
4. ***KPI reporting dashboards** should be created or modified to support the reporting of critical business services*
5. ***Alerts** should be created to automate the detection of service availability or service performance degradation related to critical business services. Alerts should be integrated with the Event Management platform and auto-ticketing in the ITSM platform enabled.*

Quality of Service (QoS) / Performance

The integrity of the solution is dependent on Quality of Service (QoS) / Performance contributions from all elements spanning the breath of this solution. Hence the description should address all the elements listed in section 2.4.

System	Describe the impacts, if any, to the SYSTEM because of the QoS requirements

Table 25: QoS / Performance

Solution should ensure all components introduced/ modified adhere to Performance KPI standard and ensure existing performance KPI is not degraded post the change for peak load as per business projection including special events like iPhone launch, New Year.

Data Backup Policy

The solution must be capable to back up and restore data with reasonable amount of timey. This capability must have tested and restore time recorded during the testing phase.

Business Continuity and Disaster/System Recovery

Describe the redundancy built into the systems (RTO/RPO) in case of system or site failure. Business Continuity, Remote Recovery, failover and fall-back scripts must be tested before changes are made to those scripts

Automation

Deployment Automation

Describe how the solution enables deployment to be triggered from any of scheduling tool including rollback steps

Operational Activities

Describe how the solution enables operational activities including shutdown and start-up to be triggered by scheduling tool and have ability to pause, reschedule and monitor from a tool

Test Automation

Describe the automated test framework used to validate Deployment and Operational Activities including Heath Checks.

Data Migration

The thresholds are for data replication- acceptable latency and fall-back process must be defined in the Solution Architecture Design. This Solution Architecture Design must also advise what to do if data falls out of sync

Patching

List exact patch version of components (application and infrastructure elements) as part of solution introduced/ modified along with its supportability from L4 vendor.

All infrastructure patches for DB, OS, Storage, Network devices including Security/vulnerability patches must be kept up to date by go live date of change/ project. Application being

introduced, customized, should have latest patch set recommended by Level 4 partner

Capacity

The integrity of the solution is dependent on capacity contributions from all elements spanning the breath of this solution. Hence the capacity description should address all the elements listed in section 2.4.1

System Name	Describe the impacts, if any, to the System because of the capacity requirements

Table 26: Capacity Description

Include the baseline growth assumptions / projection / BVM version here if available.

Based on business projections, Project should size Application / Infrastructure to ensure that a minimum 18 months capacity is available post go live.

For every new component being added, it should be possible to report on its capacity utilization via capacity management tool.

Portability

Describe how the solution can be ported to a different environment. (E.g. from physical to cloud or from one cloud environment to another)

Audit

Describe how the logs can be audited.

Service Level Agreements (SLAs)

Describe the SLAs associated with the solution

SLA name	Description
Connection & Response time	*How long will be the connection and response time*

Switchover time	*In case of failover how long will it take to divert all traffic to the standby / redundant instance of the application*
Portability	*How long will it take for the solution to be ported from one environment to another*
Incident Management	*How soon a trouble ticket will be resolved by the vendor based on the priority level*
....	

Table 27: Service level agreements

Update of Operational Databases/Processes

If your solution has changes to the operational databases, please make sure that the respective systems and processes are updated. The changes could be new application, change to existing application, removal of application etc.

System Name	Description
SSP	*Self Service Portal updated so users can request access to the solution*
UAM Process	*User Access Management updated*
PUAM	*Solution onboarded into CyberArk for Privileged User Access Management*
CMDB	*Solution logical and physical components and services offered (ideally updated dynamically).*
Active Directory	*Centralised Authentication and Authorisation used for all components*
IDM	*Solution onboarded to SailPoint Identity and Access Management for user access provisioning*
FIM	*Solution onboarded to File Integrity Monitoring system where required for PCI compliance*

PCI CDE	*PCI credit card data environment*
SIEM	*Solution onboarded into ArcSight for Security Information and Event Management*

Table 28: Operational Databases

Environments

Production Environment

It is expected that all of the changes to the Elements described in Section 1.4 and Components described Section 2.4 will be introduced in to the Production environment. If there is any variation, then describe.

Pre-Production Environment

It is expected that all of the changes to the Elements described in Section 1.4 and Components described Section 2.4 will be introduced in to the Pre-Production environment. If there is any variation, then describe.

System Integration / User Acceptance Testing Environment

It is expected that all of the changes to the Elements described in Section 1.4 and Components described Section 2.4 will be introduced in to the System Integration / User Acceptance Testing environment. If there is any variation, then describe.

Development Environment

It is expected that all of the changes to the Elements described in Section 1.4 and Components described Section 2.4 will be introduced in to the Development environment. If there is any variation, then describe.

Testing Considerations

Testing should ensure and demonstrate, solution should meet Availability, Performance and Accuracy (Success ratio) requirement of new functionality being delivered as per TPG TELECOM benchmark. And same for existing processes/ services are not degraded, if not improved upon.

For database, demonstrate all database housekeeping activities required to maintain the integrity and performance of the database must be able to run at any time without impact to supported business process/ services.

If data replication/migration is involved, describe how

(a) The replication software and link would be tested,

(b) Monitors are active on the storage arrays and links and there is no performance drop for any of the business service supported while data migration activity is on

(c) Technology Operations staff and their Managed Service Providers are familiar with and have accepted the replication monitoring tool/s.

Business Continuity, Remote Recovery, failover and fall-back scripts must be tested before changes are made to those scripts

If project/ change is introducing new Application/Infrastructure components, recovery of OS, Application, DB from backup should be tested and demonstrated to meet within business approved outage for such scenarios

Monitoring, Alerting and Error handling scenarios introduced and modified should be validated and documented as part of testing.

Error handling: This standard requires, for Business Process/ Service added/ customised, Distinct Error Code and Error Messages should be generated for every failure scenario, across Application stack and meaningful message displayed on front end for end customer consumption. Each Error should have unique code and message to be stored in respective originating application for debugging and reprocessing

If project/ change is introducing new Application/Infrastructure components, recovery of OS, Application, DB from backup should be tested and demonstrated to meet within P1 SLA timeframe.

All feature and compliance against security requirements should be tested

Operation Design Standards Exceptions

Required for ARB presentation

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Describe any of Operations Design standards that will not be met by said solution with reasoning. Current list of standards is available at here

Security Controls Compliance

Required for ARB presentation

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Required for both SO and SA

If **ALL** of the three conditions below are met for this solution, please complete **Section A**. If **One or more** conditions fail, the project needs to complete **Section B**.

Condition 1: No new System being introduced by the initiative.

Condition 2: No new integration is being introduced by the initiative.

Condition 3: The initiative is NOT a life cycle upgrade.

Section A:

I acknowledge that the solution satisfies **ALL the three conditions** above and confirm that there will be **"NO CHANGE TO EXISTING MANDATORY SECURITY CONTROLS"** with this solution.

NOTE: As part of TRA, if the GRC team identifies a possible increase to the security risk from this initiative, the project MUST complete the mandatory security controls and provide all required information to complete the TRA.

Please acknowledge YES/NO by answering as "NO CHANGE to EXISTING" and confirming that the mandatory security controls will be completed during TRA, if requested by GRC.

Section B:

Please specify how this solution meets all required Mandatory Security Controls by populating the table below.

Detail descriptions on the requirements for these controls, along with links to the respective Enterprise Security Standard, can be found on the Mandatory Security Controls confluence page.

Control No	Name of the Control	Applicability of the Control	Compliant (Yes, No or N/A)	How? (Please provide the brief explanation on how the solution supports your answer)
1	NGFWL (Next Generatio	All traffic flows must pass through a NGFW between on-prem networks or when flowing between on-prem and public cloud.	YES	No new system added as part of this project. All existing impacted systems are

	n Firewall)	TPG's AWS Public cloud logs all traffic flows at the platform level. *Does this initiative deploy a new system into a new network on-prem?* *Are all firewalls that handle traffic for this initiative, NGFW's?*		already compliant with this control. All data exchanges (e.g., Pega–Apigee–OneApp) remain within secured, already NGFW-compliant zones.
2	WAF (Web Access Firewall)	Any web application that exposes any interface externally must ONLY exist behind one of TPG's approved WAFs. *Is this initiative introducing any publicly exposed interfaces and, if so, are these interfaces protected by an approved WAF?*	YES	New new public facing application or interface added as part of the project, all existing interfaces and systems are compliant with this control.
3	DDoS Protection (Distributed Denial of Service)	All TPG-managed network subnets (on-premises, private cloud, and public cloud) must be protected by approved DDoS detection and mitigation controls. ***Is this** initiative **introducing** any publicly exposed interfaces and, if so, are these interfaces protected by an appropriate (D)DoS solution**?***	YES	No new publicly facing API interfaced added as part of the project. All OneApp impacted interfaces for offer retrieval and interaction are not public facing.
4	EDR (Endpoint Detection & Response)	*All in-scope TPG managed endpoints, including servers, workstations, and laptops, must have an approved EDR solution installed, active, and reporting to the central management platform.*	YES	No new IT systems are onboarded as part of this solution. All existing

		Are all in-scope endpoints covered by an approved EDR solution? *The EDR Standard is available **here**.*		
5	NDR (Network Detection & Response)	*All core TPG networks, including data centre, cloud, and interconnect environments, must apply approved NDR controls to provide continuous monitoring and threat detection.* *If you are deploying a new network segment, additional steps may be required.* *You can request evidence of your compliance by emailing SAE-SecurityEngineering.* *Please include your IP(s) or IP ranges in your request.*	NA	No new network segments are introduced as part of this solution.
6	SIEM (Security Incident & Event Management)	*All systems, applications, and platforms must forward relevant security logs and events to the approved SIEM platform for monitoring, alerting, and compliance reporting..* *Are all **OS, DB**, **and App** layers **of** this initiative, appropriately logging all system and security logs to the TPG **SIEM platform?***	YES	No new system added, all existing layers log to TPG SIEM
7	Access Controls	*All systems, applications, and services must enforce RBAC and integrate with TPG's AD for identity and access management.* *Further information can be found in the **user and system access control standard (Sec 23-TS)**.*	YES	No changes to the the existing Access Controls. All applications authenticate via TPG Azure AD.
8	PUAM (Privileged User	All privileged accounts across systems, applications, and infrastructure must comply with the PAM Federated Model. System	YES	As per BAU. No changes as part of this solution

	Access Management)	owners must adopt one of the approved solutions to manage privileged access in line with TPG's control objectives. *Further information can be found in the **user and system access control standard (Sec 23-TS)**.*		
9	VM (Vulnerability Management)	All TPG-managed systems, applications, and infrastructure must undergo regular vulnerability scanning, risk assessment, and remediation in line with enterprise security requirements.	YES	Vulnerbility test to be done prior to go live in align with existing release management practises across impacted systems
10	Data Security	*All data managed, processed, or stored by TPG must be handled in accordance with the Information Classification and Handling Policy. Systems and services must implement appropriate controls (e.g., encryption at rest and in transit) based on data classification.* • *Information Classification and Handling Policy (Sec-03B)* • *Cryptographic Standard (STA_01)* *Does this initiative impact a system that stores, processes, or otherwise handles Highly Sensitive Information and, if so, is that information encrypted in transit and at rest?*	YES	No new systems introduced as part of this solution. Project does not make any changes to storing of sensitive information. Controls in place for existing systems.
11	API Security	*API Security ensures that all Application Programming Interfaces are protected against unauthorised access, misuse, and data breaches. This control covers the full API lifecycle—design, deployment, maintenance, and retirement—focusing on authentication, authorisation, data*	YES	All new or updated APIs will be exposed via APIGEE and will be secured with encrypted tokens and modern TLS.

		protection, rate limiting, input validation, and monitoring. *Follow the TPG API Security Standard. A summary of its contents include:* • *Implement strong authentication and role-based access controls.* • *Protect sensitive data in transit and at rest, and prevent excessive data exposure.* • *Enforce rate limiting, input validation, and secure configurations.* • *Monitor and log all API activity to the central SIEM.* • *Maintain an up-to-date API inventory and documentation.* *Further information can be found in the API Mandatory Controls Standard (Sec-28TS)*		API logging enabled with PDC (as per control standards). TPG corporate standard API Gateway is Apigee already configured for logging and protected via WAF.
12	Network Segmentation	*All TPG environments (on-prem, private cloud, and public cloud) must implement logical or physical segmentation aligned with security zoning principles. Critical systems must be placed in restricted zones with tightly controlled access.* *Further information can be found in the Network Security Standard (Sec-25TS)*	YES	As per BAU. No changes as part of this solution
13	Application Security	*All applications built, deployed, or integrated within TPG's environment must follow secure development practices, undergo security testing, and address findings before production release. Development teams must align with TPG's Secure Development Lifecycle (SDLC) and integrate with approved security tools and processes.*	YES	Solution relies on existing vendor or internal teams for the devlopment across all impacted systems. Project will ensure that all development impacts are

		Does this initiative: • *Use a Static and Dynamic application Security Technology,* • *Appropriately handle Errors,* • *Log securely, ensuring sensitive information is not written to logs, and* • *Securely handle sessions,* *in accordance with the TPG Enterprise Application Security Policy?*		compliant with the security control.
14	Customer Security Control	*Does this initiative ensure customer-facing systems and access paths are designed and operated to protect customer access to TPG services and reduce the risk of unauthorised access, fraud, misuse, and customer harm, including secure customer authentication and authorisation, session security, monitoring of customer access activity, and protection of customer data in accordance with TPG's Data Security Standard and privacy obligations?* *Appropriate safeguards for vulnerable customers, where the risk of harm is higher, is especially important.*	YES	Solution does not make any changes to existing Customer login management. OneApp is complaint with Customer Security Control and is compliant with this security control.

Table 29: List of Mandatory Security Controls

For detail security design aspects, please check Security Design Non-Negotiable. Please call out below if any security design aspect is not achievable by this solution.

Legal and Regulatory Aspects of the Proposed Solution

Required for both SO and SA

Legal and Regulatory

Does the solution impact any area of TPG TELECOM's compliance obligations? If yes, please describe how the solution will ensure that we remain compliant to the obligation. The TPG TELECOM obligations register can be accessed here.

Compliance Area	Describe how the solution will comply to the respective compliance (if applicable)

Table 30: TPG TELECOM Compliance Obligations

PCI DSS Compliance

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Clearly specify how the solution meets the PCI Data Security Standards or if PCI compliance is Not Applicable.

Legal Interception

If your solution includes and Core network element, you need to ensure that an LI assessment has been done and if LI is impacted its solution is reflected in the LI design document

Regulatory Codes and Acts

Required for ARB presentation Back

If the solution involves a change to any system / process that supports TPG TELECOM compliance to a Telecommunication Regulatory instrument as per TPG compliance obligations, pls describe how the solution would ensure TPG TELECOM compliance to the regulatory instruments. The list of regulatory obligations are available here.

Area	Relevant Apps	Describe how the solution will comply to the respective Act / Standard (if applicable)
Number Portability	MNP/Inp Port Admin (Paradigm one) Siebel; MNP Fusion; MNP UI; MNP OCSG; ONE MNP; PCMON	

Premium Services	AAA, Siebel	
Emergency Services	MSC, GMLC (eMOLI), STP, MGW, IPIC/SGB*; NetScout	
LI Services	LI systems	
SOCI – End user Equipment/Management	ACS; Mountainview New CPE Mgmt System	
SOCI – Network/Security Operations	Netcool, NI4CC, ITSM, SIEM New Monitoring/Control System	
Legal Requirements	Siebel; Fusion; Symbio (VML & UC)	
SOCI – Cloud Hosted Systems	Mission/Business Critical Apps that are hosted on the SaaS or a new Public Cloud environment	
Customer Rights and Protection	Siebel, BRM, Tallyman, Contact Centre, ITSM; UIM(wholesale); Symbio; TCS; HOBS; ACS	

Table 32: Legal and Regulatory Impacts

Non-Production Environment (Optional)

Required for SA

Describe the components that perform roles in the delivery of the end-to-end solution in the Pre-Production Environment.

Non-Production Environment Solution Overview

Provide a Summary description of the Model environment needed, to enable the solution to be implemented on the model

Changes to Non-Production Environment Architecture

Describe the Proposed architecture of the Model environment.

Model Building blocks (including testing tools)

Clearly describe what Building Blocks will be modified, consolidated, added or removed to enable the solution to be implemented

SYSTEM Name	New or Proposed Change	Contribution / Impact Description including impacted realised capability

Table 33: NPE Component Contribution / Impact

Deployment/Migration Strategy

ℹ Required for SA

Describe how deployment/ migration would be phased to ensure no/minimum outage to supported business process/ service

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Future Considerations

ℹ Required for both SO and SA

Appendix

ℹ Optional for both SO and SA

Where necessary use this Appendix to provide additional information associated with this Solution Outline.